

# ECA1411-EMBEDDED SYSTEM

## LAB MANUAL

**Exp No:15-** EMBEDDED C PROGRAM TO CONVERT THE HEXADECIMAL DATA 0XCFH TO DECIMAL AND DISPLAY THE DIGITS ON PORTS P0, P1 AND P2

The screenshot displays an IDE interface for an embedded C program. The main window shows the source code for `EXP15.c`, which includes `<reg51.h>` and defines a `main` function. The program initializes `hexa` to `0x08` and calculates its decimal digits (units, tens, hundreds) using a loop. The digits are then displayed on ports `P0`, `P1`, and `P2` using `while(1)` loops.

The **Build Output** window shows the following messages:

```
Build started: Project: EXP11
Build target 'Target 1'
compiling EXP15.c...
linking...
Program Size: data=9.0 xdata=0 code=37
creating hex file from ".\Objects\EXP11"...
".\Objects\EXP11" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:00
```

The **Execution** window shows the program running. The **Parallel Port 2** window displays the value `0` on `P2`. The **Parallel Port 1** window displays the value `8` on `P1`. The **Parallel Port 0** window displays the value `0` on `P0`.

The **Watch** window shows the following variables and their values:

| Name               | Value | Type |
|--------------------|-------|------|
| <Enter expression> |       |      |

The status bar at the bottom indicates the simulation is running, with a time of `t1: 0.00043186 sec` and a clock frequency of `L23 C1`.